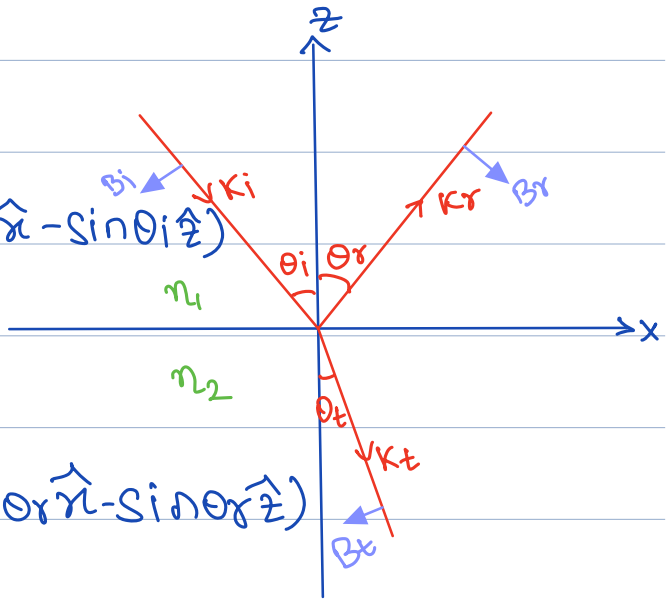
Problem 9.17 Analyze the case of polarization *perpendicular* to the plane of incidence (i.e. electric fields in the y direction, in Fig. 9.15). Impose the boundary conditions (Eq. 9.101), and obtain the Fresnel equations for $\tilde{E}_{0_R}$ and $\tilde{E}_{0_T}$. Sketch $(\tilde{E}_{0_R}/\tilde{E}_{0_I})$ and $(\tilde{E}_{0_T}/\tilde{E}_{0_I})$ as functions of θ_I, for the case $\beta = n_2/n_1 = 1.5$. (Note that for this β the reflected wave is *always* 180° out of phase.) Show that there is no Brewster's angle for *any* n_1 and n_2: $\tilde{E}_{0_R}$ is *never* zero (unless, of course, $n_1 = n_2$ and $\mu_1 = \mu_2$, in which case the two media are optically indistinguishable). Confirm that your Fresnel equations reduce to the proper forms at normal incidence. Compute the reflection and transmission coefficients, and check that they add up to 1.



$$E_i = E_{0i}\, e^{i(\bar{k}_i \cdot \bar{r} - \omega t)}\, \hat{y}$$

$$B_i = \frac{E_{0i}}{v_1}\, e^{i(\bar{k}_i \cdot \bar{r} - \omega t)} (\cos\theta_i \hat{x} - \sin\theta_i \hat{z})$$

$$E_r = E_{0r}\, e^{i(\bar{k}_r \cdot \bar{r} - \omega t)}\, \hat{y}$$

$$B_r = \frac{E_{0r}}{v_1}\, e^{i(\bar{k}_r \cdot \bar{r} - \omega t)} (\cos\theta_r \hat{x} - \sin\theta_r \hat{z})$$

$$E_T = E_{0T}\, e^{i(\bar{k}_T \cdot \bar{r} - \omega t)}\, \hat{y}$$

$$B_T = \frac{E_{0T}}{v_2}\, e^{i(\bar{k}_T \cdot \bar{r} - \omega t)} (\cos\theta_T \hat{x} - \sin\theta_T \hat{z})$$

$$\epsilon_1 E_{1\perp} = \epsilon_2 E_{2\perp} \;;\; E_{1\parallel} = E_{2\parallel} \;;\; B_{1\perp} = B_{2\perp} \;;\; \frac{B_{1\parallel}}{\mu_1} = \frac{B_{2\parallel}}{\mu_2}$$

$$\theta_i = \theta_r \;;\; \frac{\sin\theta_i}{\sin\theta_T} = \frac{n_2}{n_1} = \frac{v_1}{v_2}$$

$$E_1^{\parallel} = E_2^{\parallel} \Rightarrow E_{0i} + E_{0R} = E_{0T} \quad -(1)$$

$$B_1^{\perp} = B_2^{\perp} \Rightarrow \frac{E_{0i}\sin\theta_i}{v_1} + \frac{E_{0R}\sin\theta_i}{v_1} = \frac{E_{0T}\sin\theta_T}{v_2} \quad \text{same as } (1)$$

$$\frac{B_1''}{\mu_1} = \frac{B_2''}{\mu_2} \Rightarrow \frac{1}{\mu_1}\left[-\frac{E_{0i}}{v_1}\cos\theta_i + \frac{E_{0R}}{v_1}\cos\theta_i\right] = \frac{1}{\mu_2}\left[\frac{E_{0T}}{v_2}\cos\theta_T\right]$$

$$\Rightarrow E_{0i} - E_{0R} = \frac{\mu_1 v_1}{\mu_2 v_2}\frac{\cos\theta_T}{\cos\theta_i} E_{0T}$$

$$\frac{E_{0T}}{E_{0i}} = \frac{2\cos\theta_i \mu_2 v_2}{\cos\theta_i \mu_2 v_2 + \cos\theta_T \mu_1 v_1} \qquad \frac{E_{0R}}{E_{0i}} = \frac{\cos\theta_i \mu_2 v_2 - \cos\theta_T \mu_1 v_1}{\cos\theta_i \mu_2 v_2 + \cos\theta_T \mu_1 v_1}$$

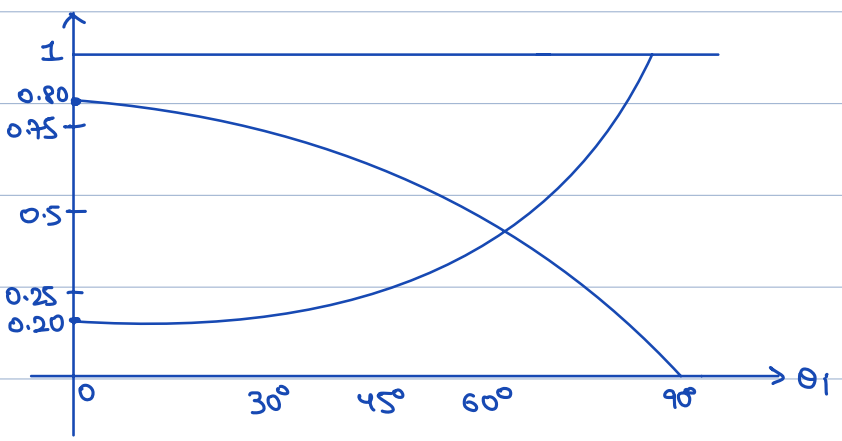
$$\frac{v_1}{v_2} = 1.5, \quad \mu_1 \cong \mu_2$$

$$\Rightarrow \frac{E_{0T}}{E_{0i}} = \frac{2\cos\theta_i}{\cos\theta_i + \frac{3}{2}\cos\theta_t} \qquad \frac{\sin\theta_i}{\sin\theta_T} = 1.5$$

$$9/4\cos^2\theta_i = 1 - \cos^2\theta_t$$

$$= \frac{2\cos\theta_i}{\cos\theta_i + \sqrt{9/4 - \cos^2\theta_i}}$$

$$\theta_i = 0 \Rightarrow \frac{2v_2}{v_1 + v_2} = \frac{4}{5}$$



Brewster angle $\Rightarrow E_{0R} = 0$

$$\Rightarrow \frac{\cos\theta_i}{\cos\theta_T} = \frac{\sin\theta_i}{\sin\theta_T} = \frac{v_1}{v_2} = c$$

$$\Rightarrow \cos^2\theta_i = c^2\cos^2\theta_T$$

$$\Rightarrow 1 - \sin^2\theta_i = c^2 - c^2\sin^2\theta_T = c^2 - \sin^2\theta_i$$

$\Rightarrow$ There exist a brewster angle iff $c = 1$

i.e $v_1 = v_2 \Rightarrow$ both the media are optically indistinguishable

At normal forms: $\theta_i = \theta_T = 0^\circ$

$$R = \left(\frac{E_{0R}}{E_{0i}}\right)^2 \qquad T = \frac{\epsilon_2 v_2}{\epsilon_1 v_1}\frac{\cos\theta_t}{\cos\theta_i}\left(\frac{E_{0T}}{E_{0i}}\right)^2$$

$$R + T = \frac{(\cos\theta_i \mu_2 v_2 - \cos\theta_T \mu_1 v_1)^2 + (2\cos\theta_i \mu_2 v_2)^2 \frac{\cos\theta_T}{\cos\theta_i}\frac{\epsilon_2 v_2}{\epsilon_1 v_1}}{(\cos\theta_i \mu_2 v_2 + \cos\theta_T \mu_1 v_1)^2}$$

$$= 1$$